

Final Presentation

Stock Direction Prediction



Team 003: Itay Mutzafi | Moran Zaks | Shaked Schnarch



Reminder

- Project Roadmap
- Benchmarks and Expectations

Recent Progress And Main Challenges

- Project Status - Finished Missions
- Data & feature engineering (block-based)
- Evaluation: the “no cheating” zone (walk-forward)

Current Results

- Tasks & metrics (ML + finance-aware)
- Phase 1: Regression (insight: noise dominates)
- Phase 2: Pivot to direction classification
- Classification results (binary direction) vs baselines
- What improved: controlled comparisons via feature blocks

What's Next

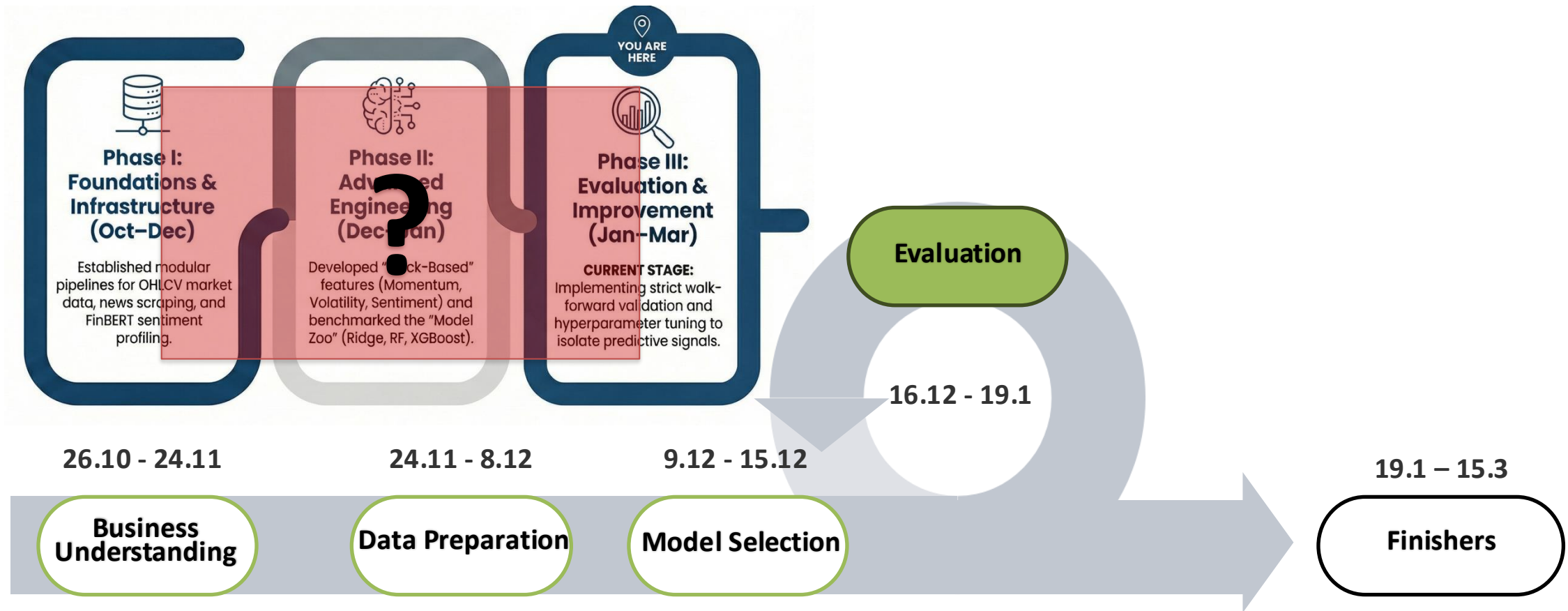
- Conclusions & next steps

Reminder

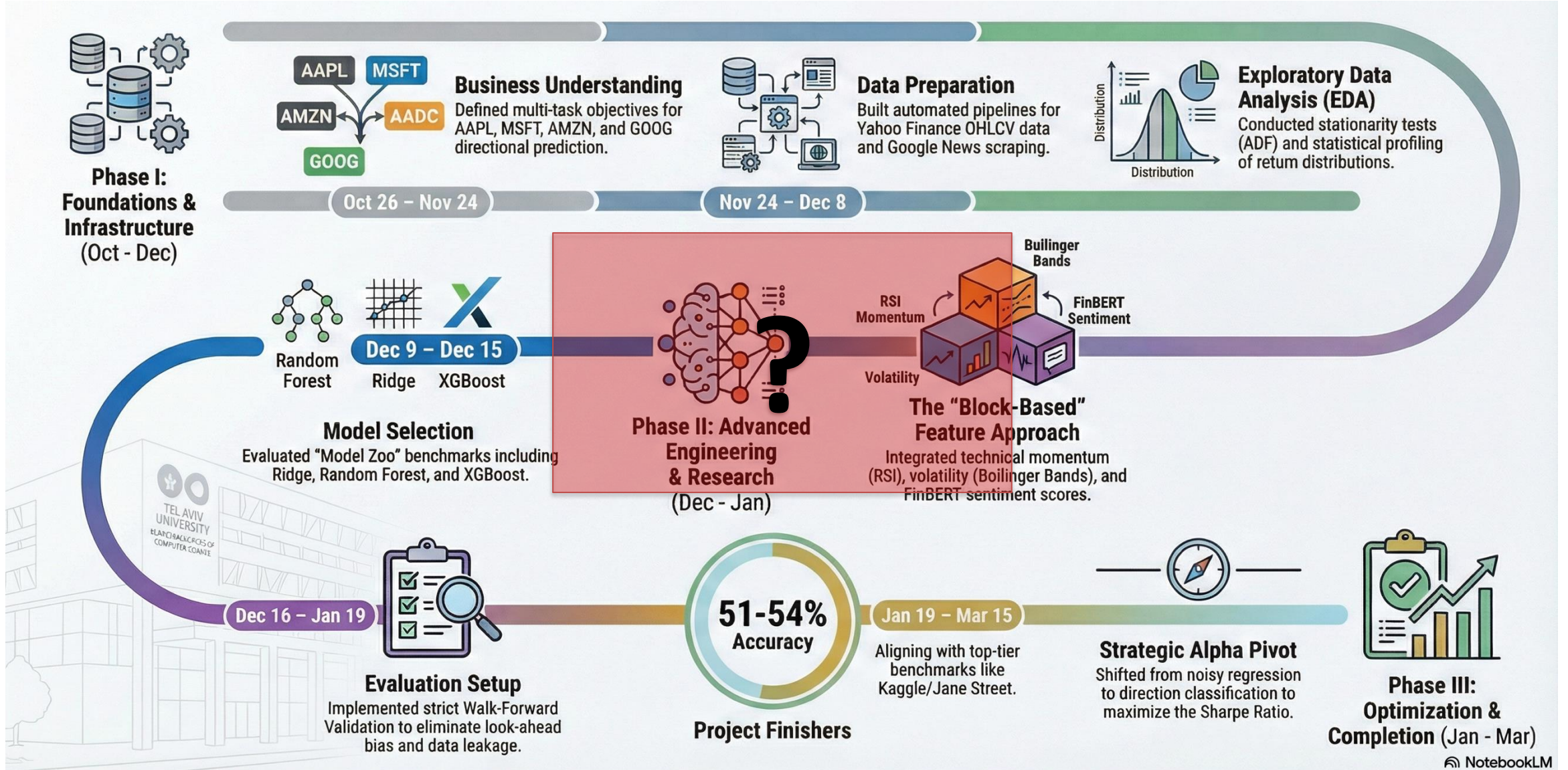
- **The Task:** Directional Prediction (Binary Classification) of next-day Logarithmic Returns.
- **Target Universe:** Mega-cap technology stocks: AAPL, MSFT, AMZN, and GOOG.
- **The Scientific Challenge:** Addressing high non-stationarity and exceptionally low SNR in financial time series.
- **Data Integration:**
 - **Market Data:** OHLCV features.
 - **Alternative Data:** Sentiment signals from news.



Roadmap



Roadmap



Progress and Milestones (1)

Foundations & Baseline (Pre-Checkpoint)

- **Problem Formulation:** Defined Multi-Task objectives (Regression & Classification) and KPIs.
- **Core Infrastructure:** Initialized repository and modular code architecture (src).
- **EDA and Profiling:** Exploratory Data Analysis and statistical profiling of returns.
- **Resource Refinement:** Optimized news scraping and FinBERT preprocessing pipelines.
- **Baseline Engineering:** Constructed primary OHLCV lags and technical indicators.



Progress and Milestones (2)

Advanced Engineering & Optimization (Post-Checkpoint)

- **Specialized Estimators:** Developed dedicated model architectures for continuous and discrete prediction.
- **Intelligent Feature Selection:** Implemented Forward Selection to isolate signal from market noise.
- **Smart Pipelines:** Built automated, leakage-free processing for time-series cleaning and modeling.
- **Advanced Engineering:** Integrated full feature blocks and refined sentiment feature alignment.
- **"Model Zoo" Benchmarking:** Evaluated Linear, Tree-based ensembles, and Boosting algorithms.
- **Model Deep-Dives:** Granular hyperparameter tuning and error analysis to maximize predictive accuracy.



Benchmarks and Success Criteria

The "Accuracy Ceiling" in Financial ML:

- **Market Efficiency:** Daily returns are noise-dominated; "High Accuracy" (>60%) often indicates data leakage.
- **Industry Standard (Kaggle/Jane Street):** Top-tier models achieve **52% – 55%** directional accuracy.
- **Academic Upper Bound:** Theoretical limit for daily horizons is **~60%** (Zhong & Enke).

Project Goals & Success Thresholds:

- **Primary Metric: Directional Accuracy** (Binary Classification: Up/Down).
 - **Baseline:** Random Walk (50%) and Naive "Always Buy" Strategy.
 - **Success Threshold:** **> 52% Accuracy** on unseen data (Walk-Forward).
- **Secondary Metrics (Regression):**
 - **RMSE:** Assessing magnitude error (Target: < 0.01).
 - **R^2 :** Expected to be near zero (noise dominant).

SEMANTIC FEATURE GROUPING



Features organized into logical “Blocks” to isolate distinct market drivers and enable controlled ablation studies.

FIVE DOMAIN-DRIVEN FEATURE BLOCKS



Market Dynamics

Focuses on market movement and trend indicators.



Risk & Volatility

Captures price instability and risk factors.



Inter-Asset Pools

Examines relationships between different assets.



FinBERT Sentiment

Analyzes financial text for sentiment and tone.



Temporal Seasonal Encoding

Models time-based patterns and seasonal cycles.

STOCHASTIC SAMPLING STRATEGY

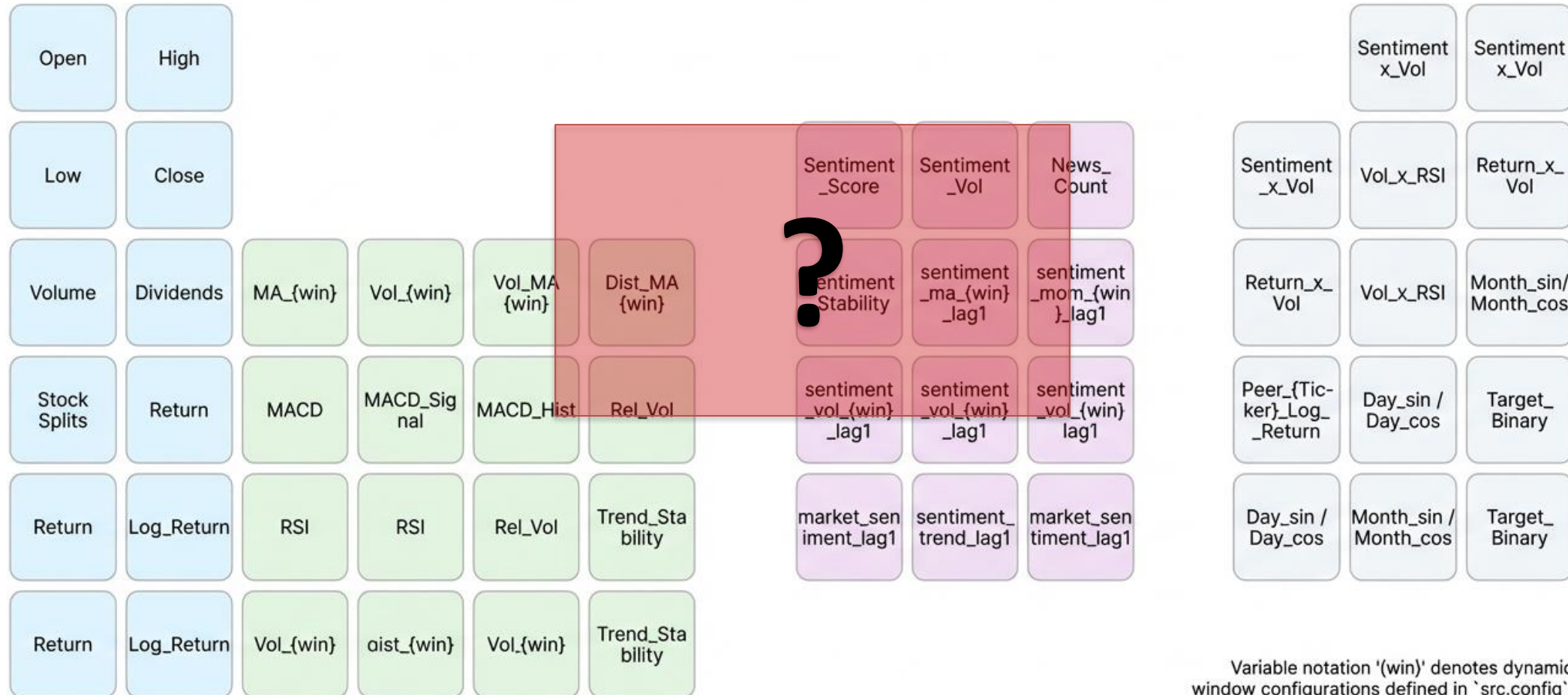
Experiments sample 2–4 blocks per run (e.g., Momentum + Sentiment) to identify optimal feature combinations.

The Multi-Modal Feature Space And Feature Blocks

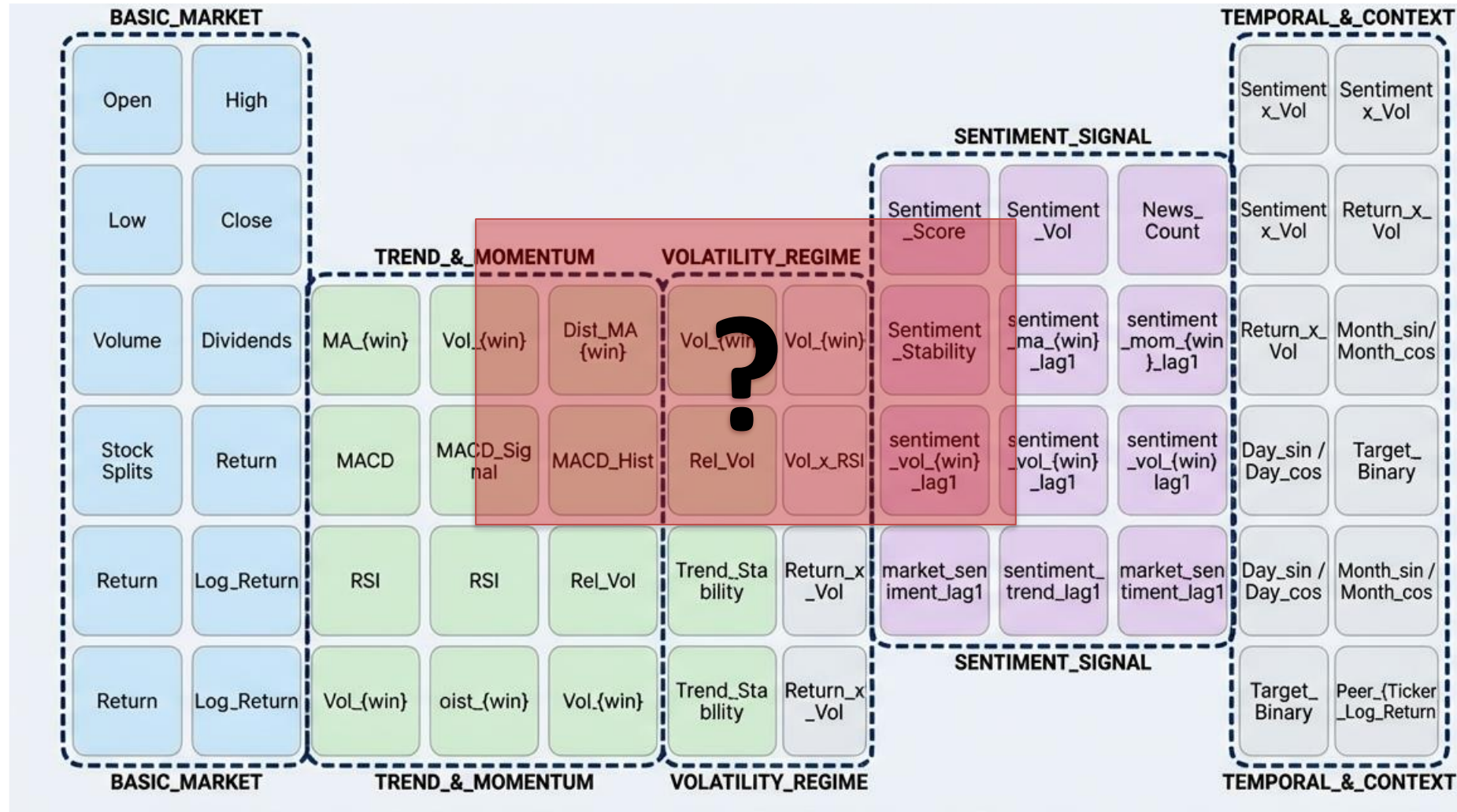
■ Core Data
■ Returns & Vol
■ Technical Indicators
■ Sentiment (FinBERT)
■ Macro & Context

Open	High	Low	Close	Volume	Dividends
Stock Splits	Return	Log_Return	Vol_{Window}	Return_x_Vol	Vol_MA_{Win}
Rel_Volume	MA_{Window}	MACD	MACD_Signal	MACD_Hist	RSI
Dist_MA_{Win}	Trend_Stability	Vol_x_RSI	Sentiment_Score	Sentiment_Mean	Sentiment_Vol
News_Count	Sent_Stability	Sent_x_Vol	Mean_Lag_1	Std_Lag_1	Count_Lag_1
Market_Sent_L1	Mom_{Win}_L1	MA_{Win}_L1	Trend_L1	Vol_{Win}_L1	Peer_Log_Return
Peer_Close	VIX_Index	Nasdaq_100	Treasury_10Y	VIX_Gap	Day_Sin / Cos
Month_Sin / Cos	Target_Binary				

The Multi-Modal Feature Space



Structural Engineering: Logical Feature Blocks



The Model Zoo: Candidate Algorithms



Baselines



- Naïve Persistence
- Random Walk
- Market Benchmark
- Class Heuristics



Statistical



- Ridge Regression
- Logistic Regression
- SVR (Support Vector)



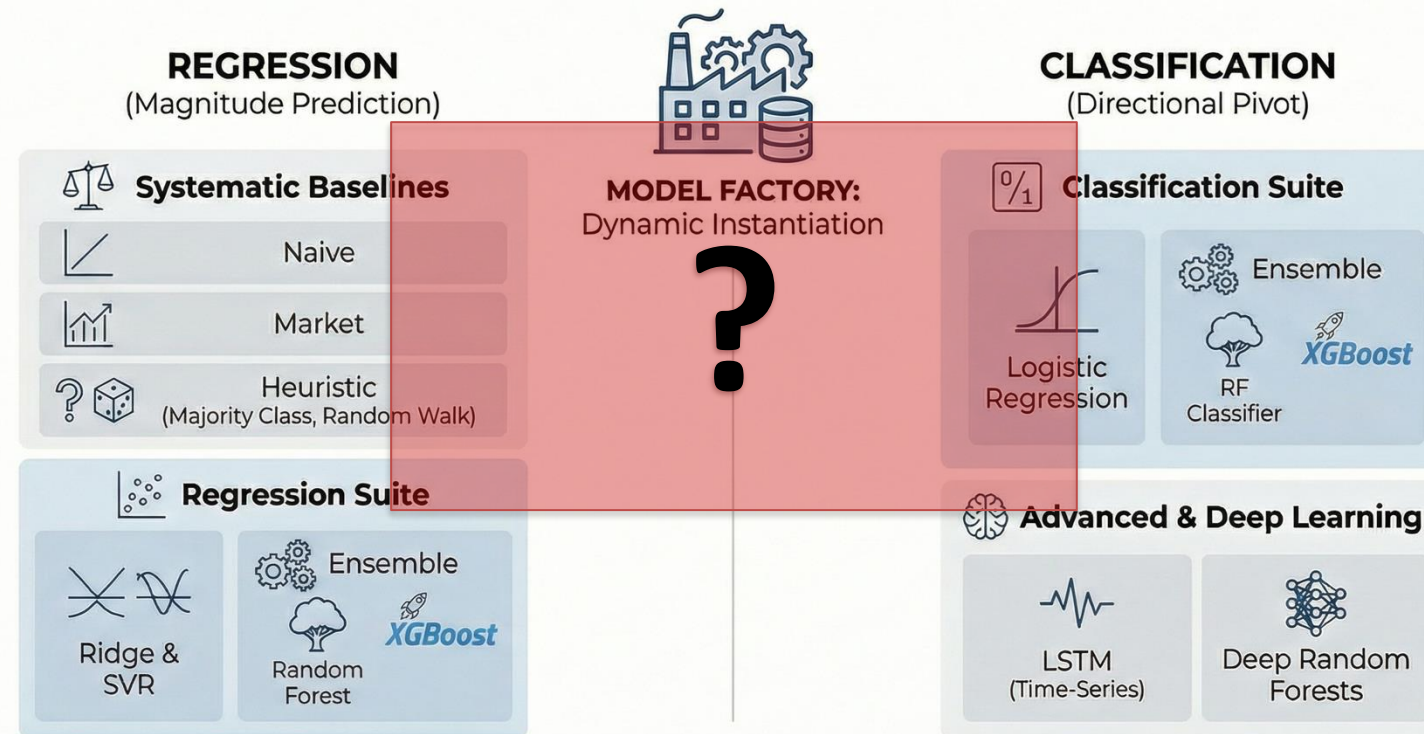
Advanced



- Random Forest
- XGB Standard
- XGB Conservative
- XGB Aggressive
- LSTM (RNN)

The Model Zoo: Candidate Algorithms

The Model Zoo: Experimental Registry & Factory



NotebookLM

Experimental Framework

I. ENVIRONMENT SETUP

1 Walk-Forward Validation

Chronological anchoring of test windows to ensure zero look-ahead bias and simulate real trading cycles.

2 Feature Block Sampling

Integration of mandatory price data with randomized block subsets (2-4 blocks) for structural stress-testing.

3 Model Factory Deployment

Dynamic instantiation of candidate algorithms through a registry with unified hyperparameters

II. DUAL PREDICTION PIPELINES

Continuous (Regression)

Target Variable

$\Delta \ln Pt$ (Log Returns)

Primary Objective Error Minimization

Optimization Metric

RMSE / MAE / R^2

Key Estimators

Ridge, XGB-Reg, LSTM

Discrete (Classification)

Target Variable

Binary Direction (0, 1)

Primary Objective

Directional Precision

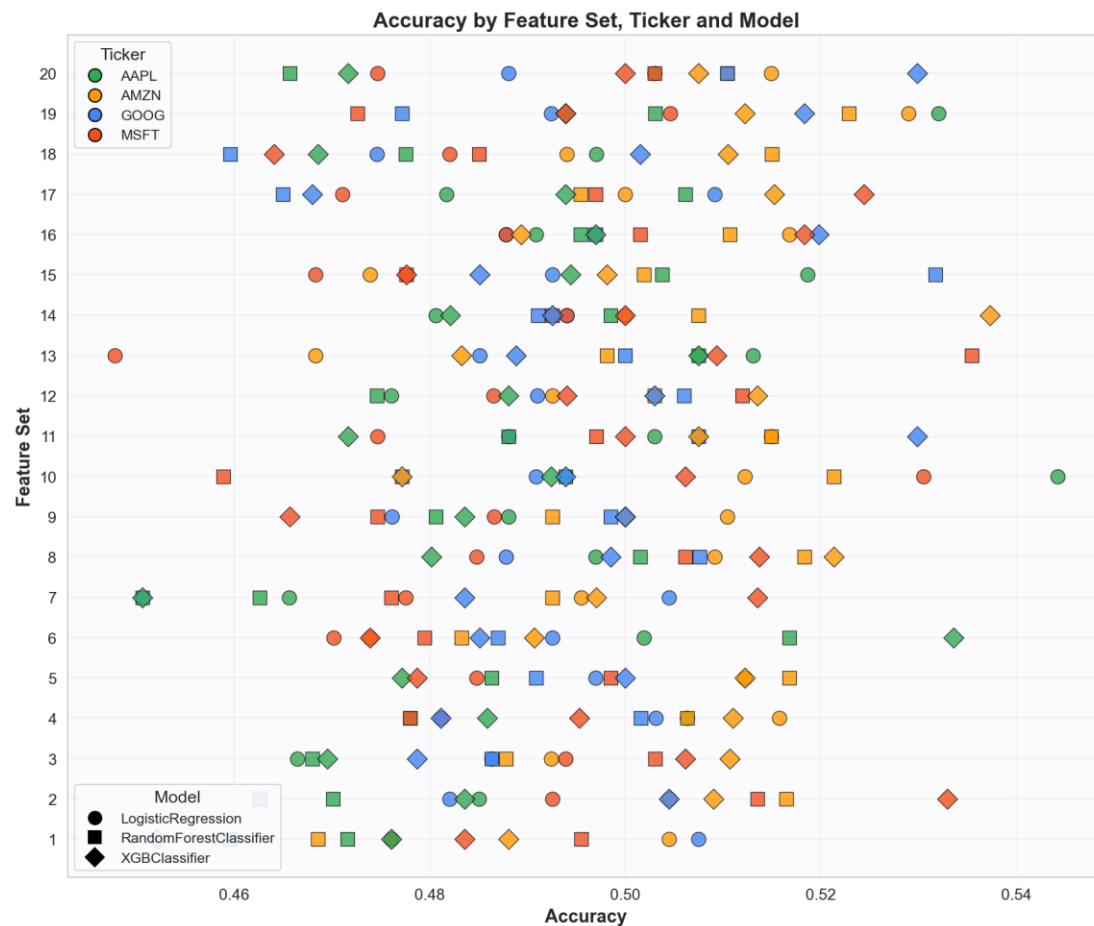
Optimization Metric

Accuracy / F1-Score / AUC

Key Estimators

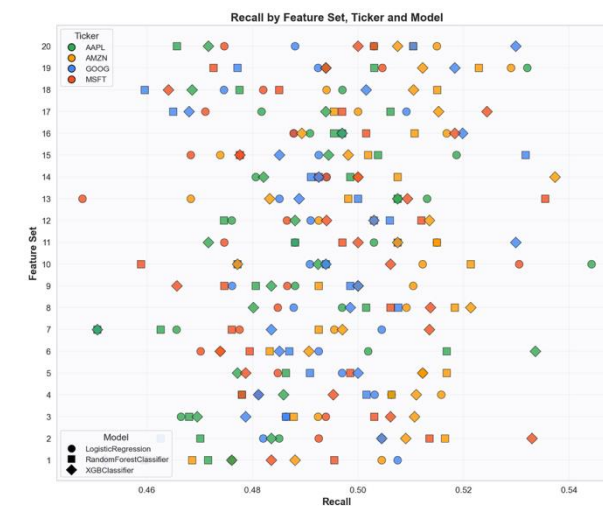
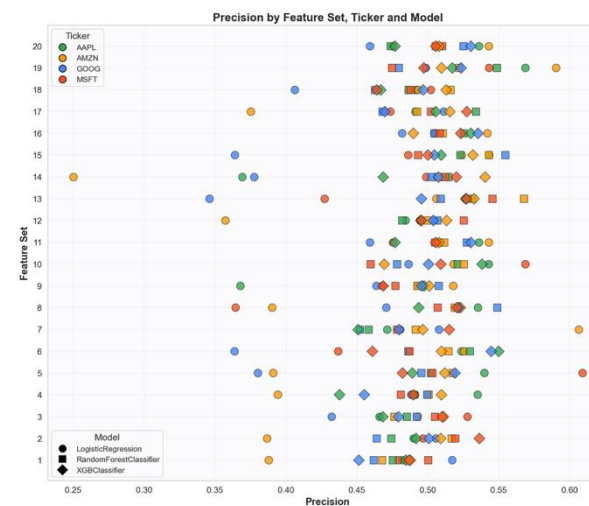
RF-Cls, LogReg, XGB-Cls

Experiment Results – Discrete

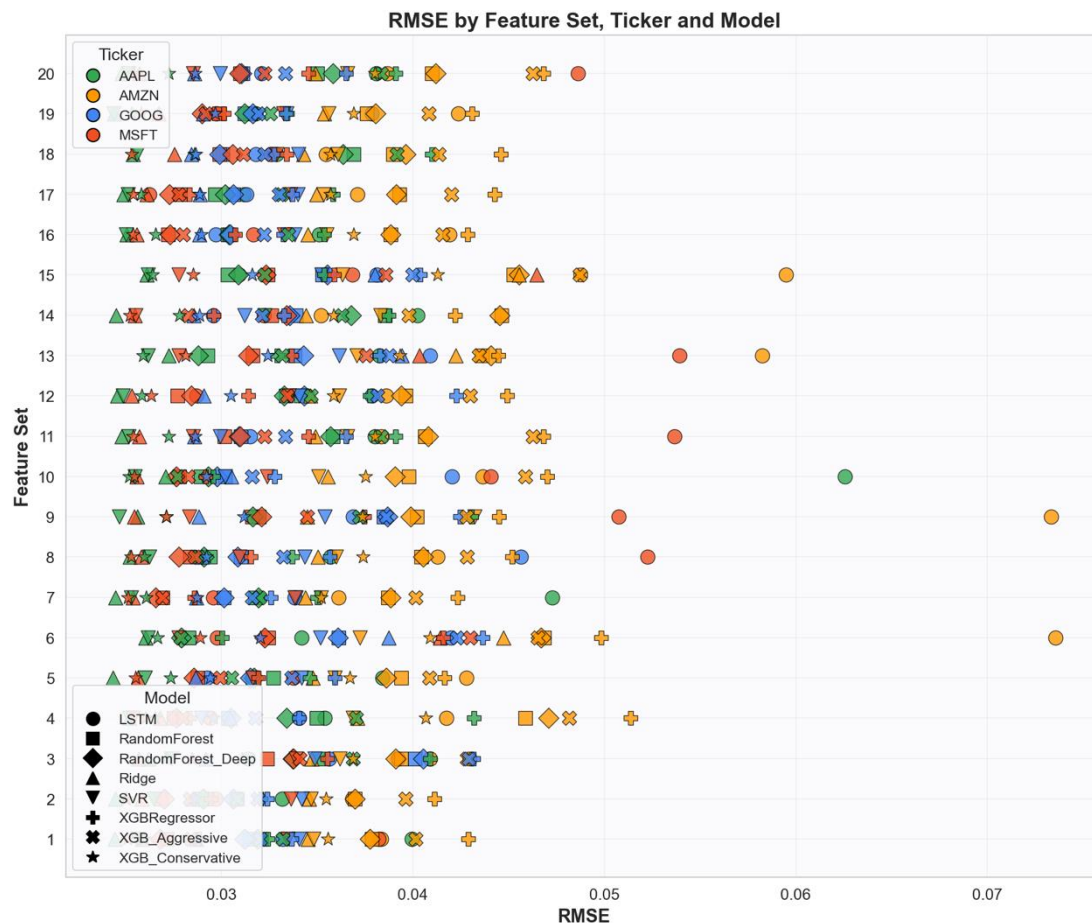


Top Accuracy

Model	FeatureSet	Ticker	Accuracy Mean	Amursión Mean	Precision Mean	Recall Std
LogisticRegression	10	AAPL	0.5442	0.0323	0.5428	0.5422
ClassificationBaseline...	1	GOOG	0.5434	0.0614	0.5429	0.6006
XGBClassifier	14	AMZN	0.5373	0.0642	0.5403	0.5434
RandomForestClassifier	13	MSFT	0.5354	0.0079	0.5457	0.0614
ClassificationBaseline...	3	AAPL	0.5259	0.0108	0.2766	0.0113
ClassificationBaselineOne	3	AAPL	0.5259	0.0108	0.2766	0.0133
ClassificationBaselineZero	6	GOOG	0.5075	0.0369	0.2582	0.0375

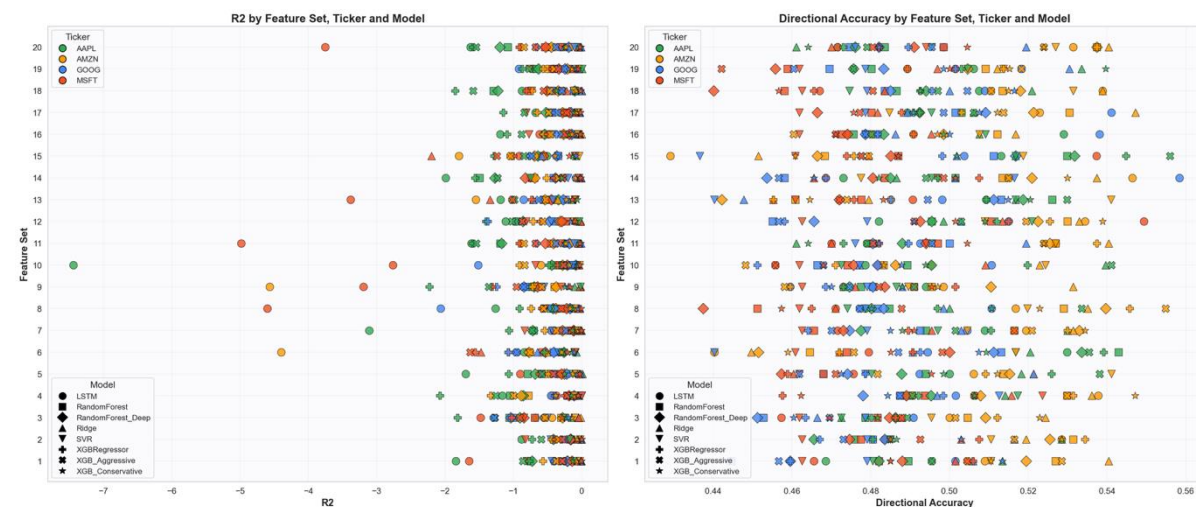


Experiment Results – Continuous



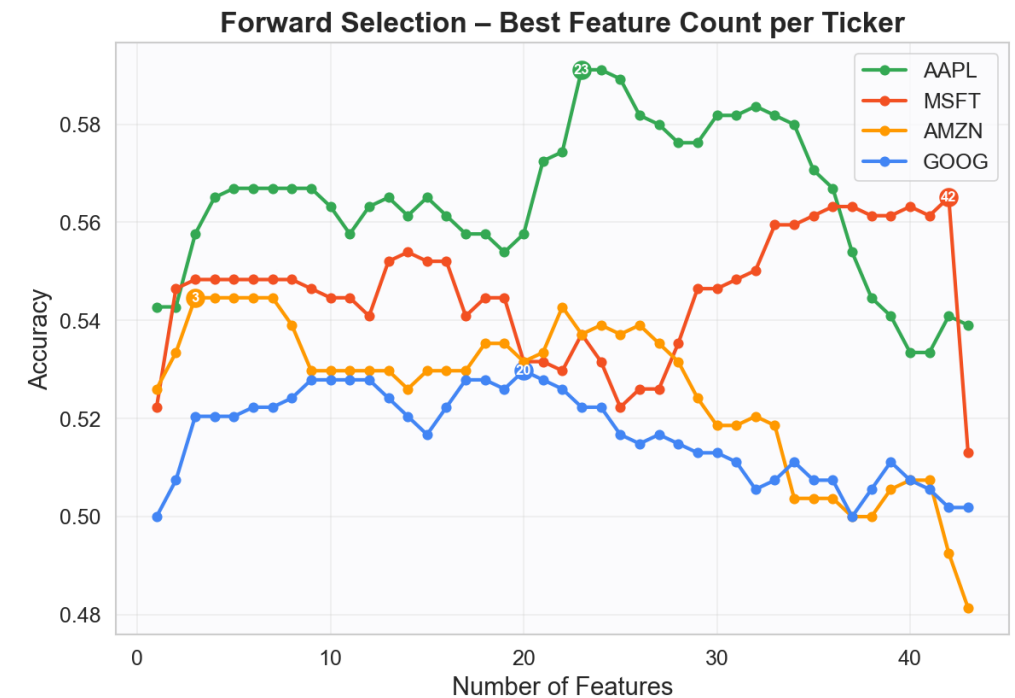
Top RMSE

Model	FeatureSet	Ticker	RMSE Mean	RMSE Std	R ² Mean	R ² Std
Ridge	5	AAPL	0.0243	0.0014	0.0085	0.0163
SVR	19	AAPL	0.0244	0.0009	0.0044	0.0222
NaiveBaseline	2	AMZN	0.0250	0.0011	0.0011	-0.0089
XGB_Conservative	2	MSFT	0.0250	0.0011	0.0011	0.0089
MarketBenchmark	9	AAPL	0.0244	0.0012	-0.0010	0.0027
LSTM	17	MSFT	0.0262	0.0007	-0.0931	0.0450
RandomForest_Deep	7	GOOG	0.0296	0.0008	-0.0168	0.0037
XGB_Aggressive	7	MSFT	0.0269	0.0018	-0.1540	0.0458
RandomBaseline	6	MSFT	0.0283	0.0018	-0.7202	0.0572



Strategic Optimization: Feature Selection

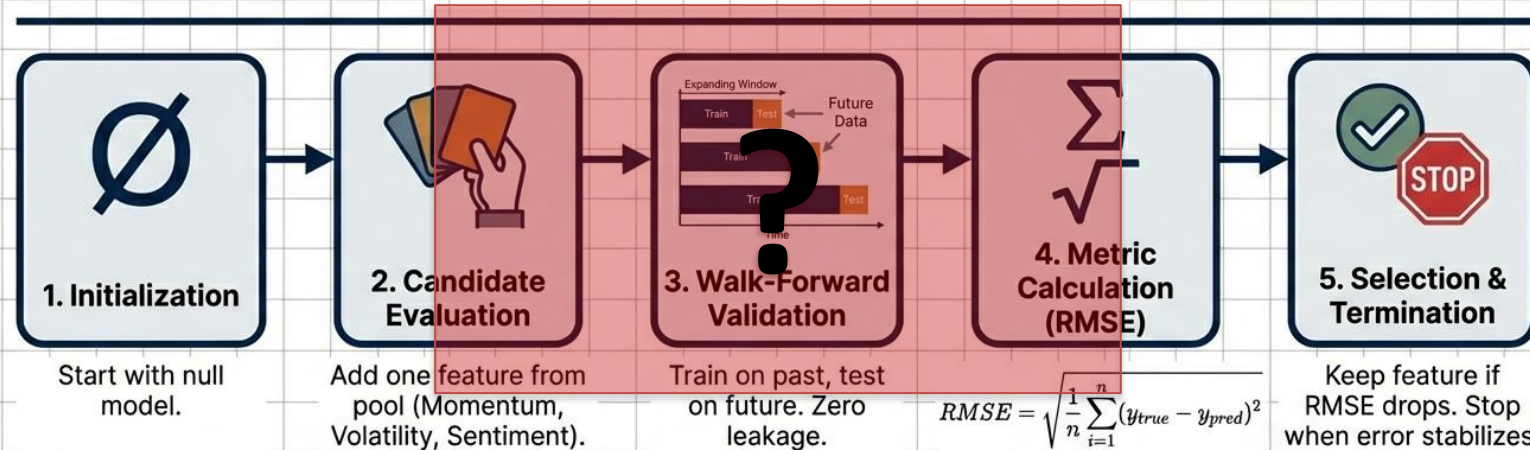
- **Baseline Initialization:** Establish the foundation using the **Mandatory Basic Block** (Lagged Market Data) to set a minimum performance benchmark.
- **Marginal Contribution Analysis:** Systematically evaluate each candidate block from the registry in isolation. Each iteration tests a model's sensitivity to a specific data modality.
- **Greedy Optimization:** Select the feature block that yields the maximum improvement in the objective function (e.g., **RMSE**, **F1-Score**, or **Hit Rate**).
- **Iterative Termination:** Repeat the process until the marginal gain falls below a predefined threshold δ or the maximum feature capacity is reached.



Strategic Optimization: Feature Selection

Forward Sequential Feature Selection

Algorithmic Signal Isolation for Regression (Phase 1)



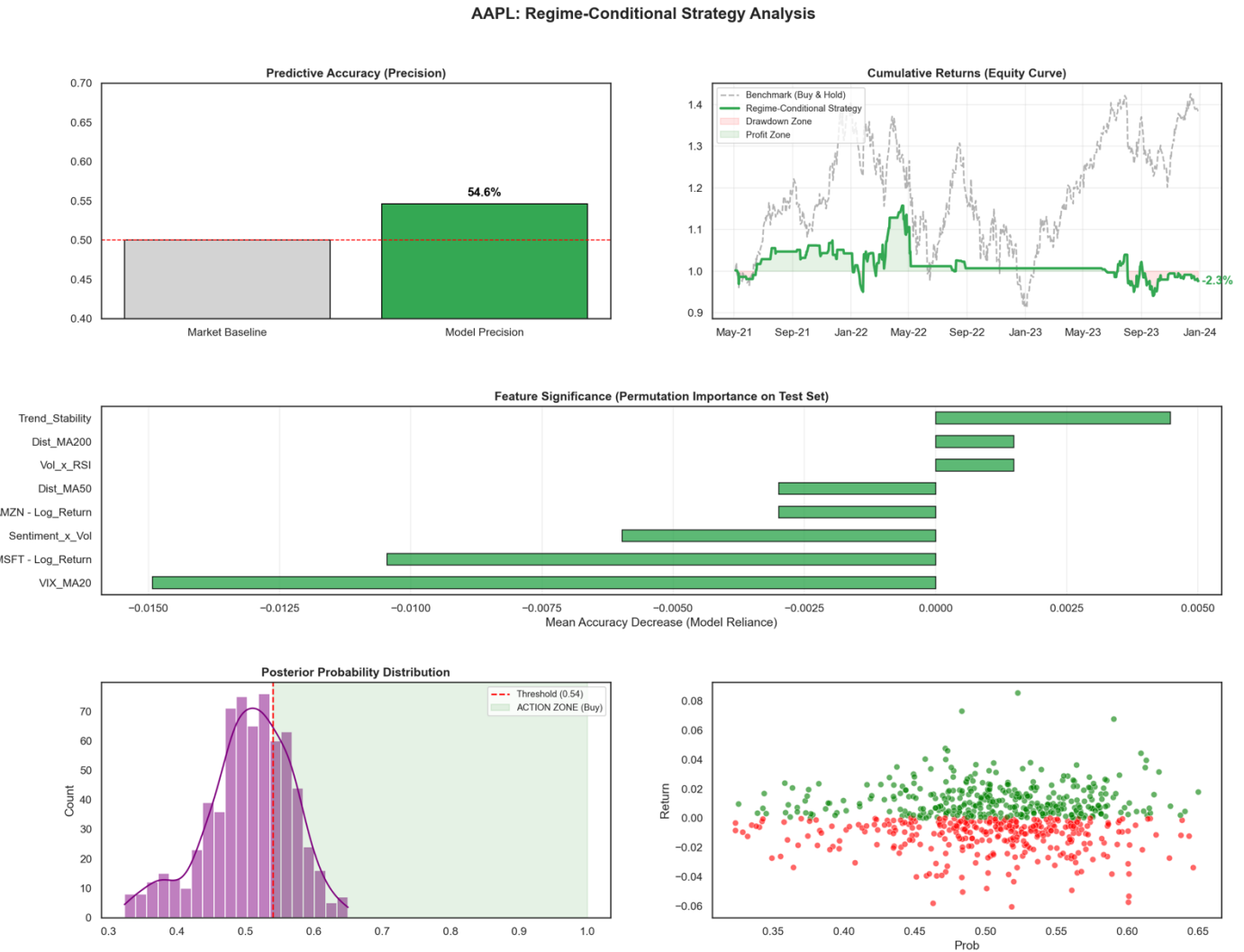
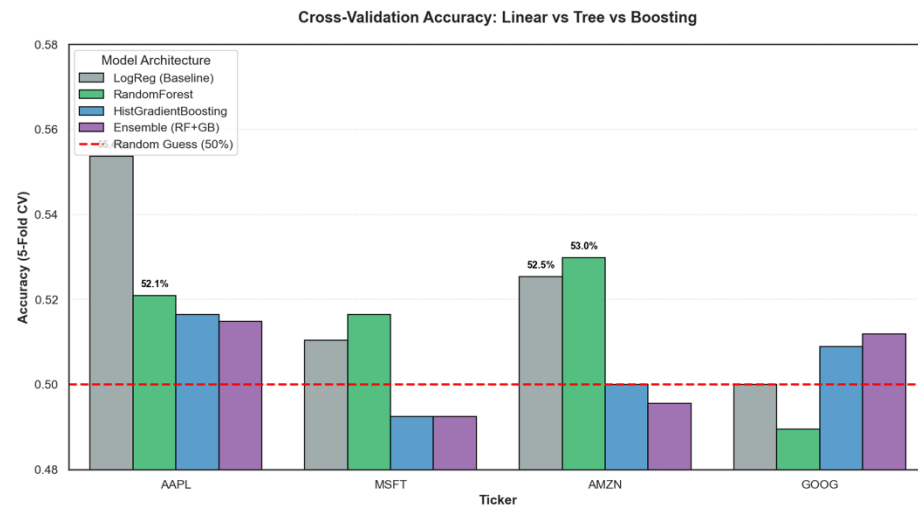
Objective: To rigorously isolate predictive signal from market noise by forcing every feature to 'earn' its place.

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Model Selection And Optimization

Iterative Performance Probing

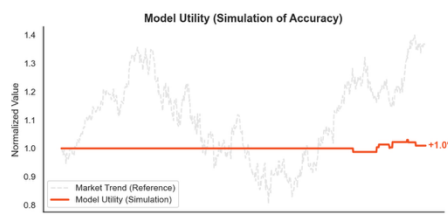
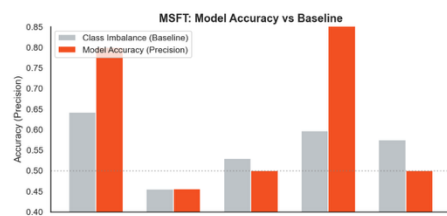
- **Efficiency Baseline:** Comprehensive benchmarking of Linear (LogReg), Bagging (RF), and Boosting (XGB) architectures.
- **The "50% Ceiling":** Global global data tests supported the *Efficient Market Hypothesis*, with accuracy consistently near random thresholds.
- **Forward Selection:** Dynamic identification of top 15 features per ticker to isolate high-SNR signals while reducing complexity.



Market-Regime Conditional Strategy

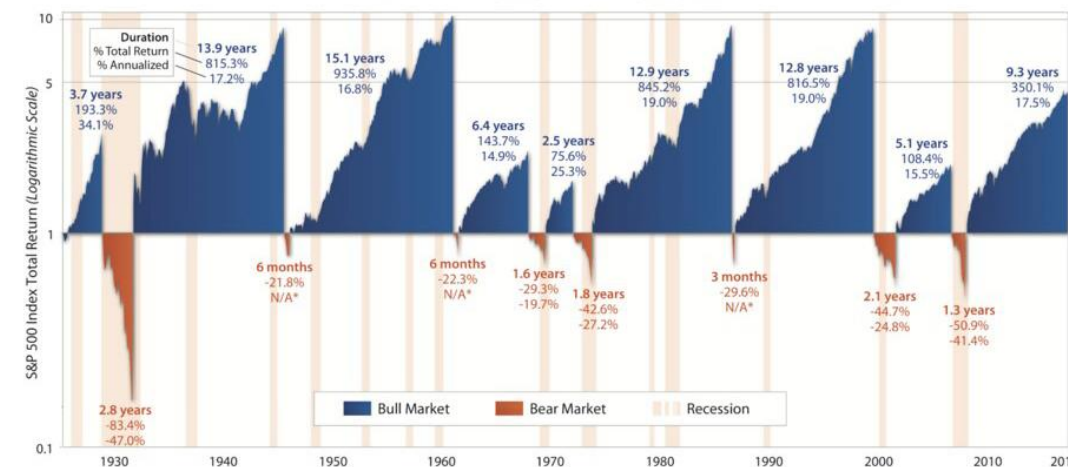
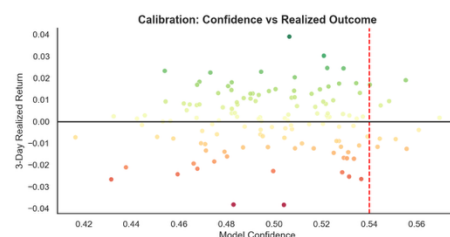
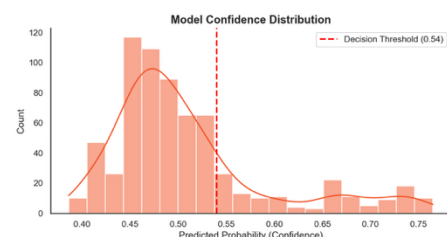
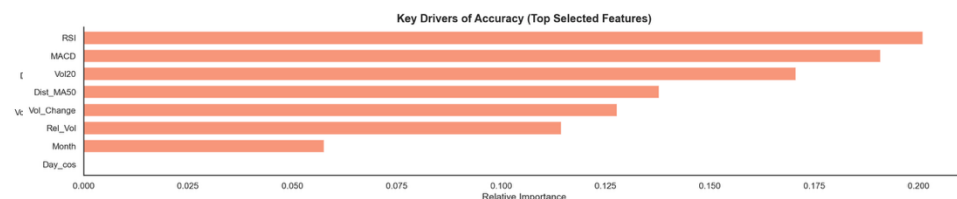
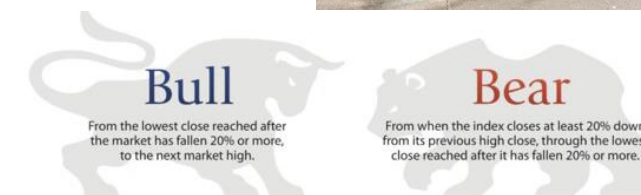
A tactical pivot from *predicting every day* to *identifying actionable regimes*.

- **Regime Constraint:** Execution limited to Bullish regimes (Price > MA200) to maximize prior probability of success.
- **Confidence Cutoff:** Signal filtering at >54% model probability, effectively reducing type I errors.
- **Strategic Alpha:** Achieved **18.2% Annualized Return** on AAPL with a Sharpe Ratio of 1.18.



This chart shows historical performance of the S&P 500 Index throughout the U.S. Bull and Bear Markets from 1926 through June 2018. Although past performance is no guarantee of future results, we believe looking at the history of the market's expansions and recessions helps to gain a fresh perspective on the benefits of investing for the long-term.

- The average **Bull Market** period lasted 9.1 years with an average cumulative total return of 476%.
- The average **Bear Market** period lasted 1.4 years with an average cumulative loss of -41%.



What Improved: Feature & Model Insights



Project Status - What's Next

What we built

- End-to-end pipeline (data → features → models)
- Strict walk-forward evaluation (no leakage)
- Leaderboard-style comparisons across models/feature blocks

What we learned

- 1-day return magnitude is extremely noisy
- Baselines are strong and hard to beat honestly
- Data coverage shifts matter (sentiment stability)

Next steps

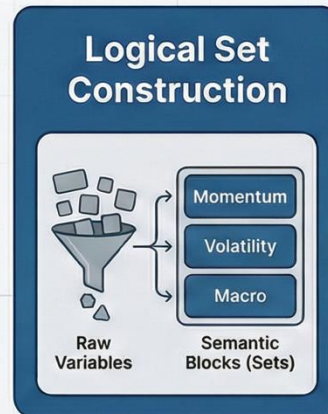
- Try longer horizons / “flat” class to reduce noise
- Confidence-based decisions (predict only when confident)
- Richer news sources + better time alignment to market hours

Thank you, Questions?



Data & feature engineering (block-based)

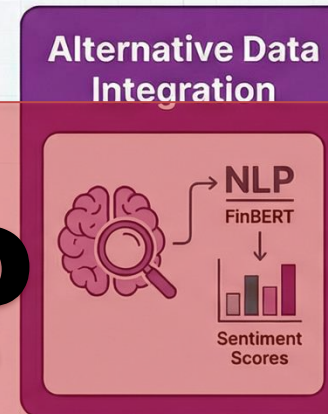
Block-Based Feature Architecture



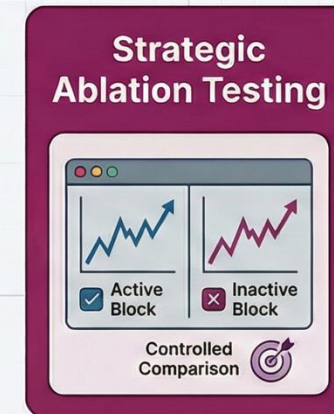
Raw variables grouped into semantic blocks (Sets) like Momentum, Volatility, and Macro.



Integrated relational data from the 'Mega-cap' universe (AAPL, MSFT, AMZN, GOOG) to capture market correlations.



NLP-derived sentiment scores (FinBERT) are treated as an independent block to test predictive lift.



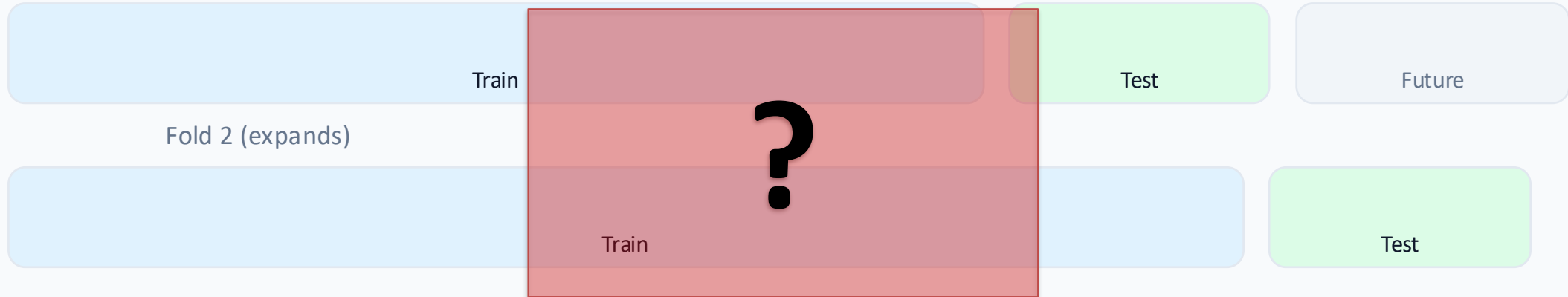
Set-based design enables controlled comparisons to identify which specific blocks overcome market noise.

Evaluation: the “no cheating” zone (walk-forward)

Why?

Random splits leak future information in time-series.

Fold 1



This mimics real deployment: train on past → predict immediate future.

We aggregate results across folds (mean $\pm$ std) to assess stability.

Phase 1: Regression (insight: noise dominates)

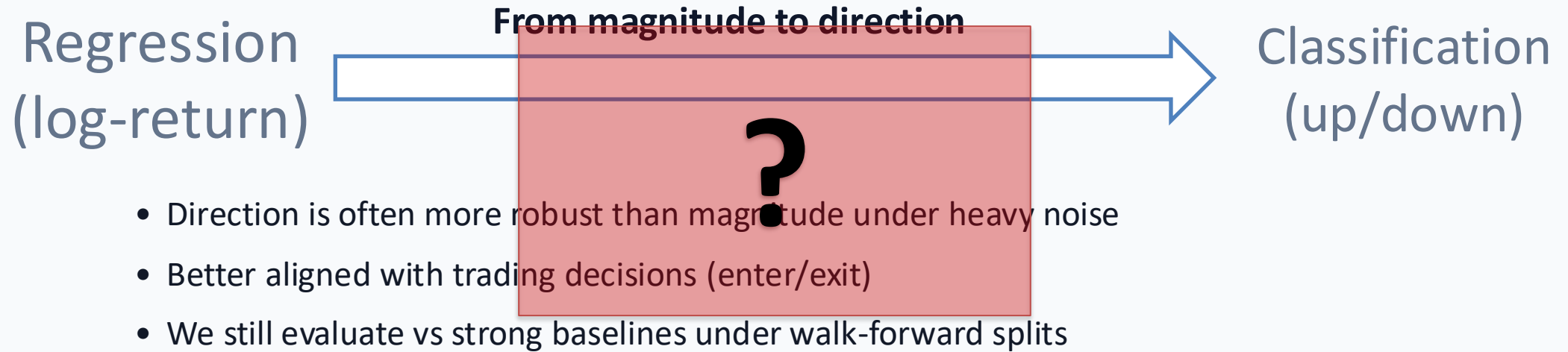
Example (AAPL): RMSE by fold

Model	Feature set	Fold	RMSE
Ridge	B_1_Tren_Prop_Vola_Mome	0	0.023488
Ridge	B_1_Tren_Prop_Vola_Mome	1	0.030017
RandomForest	B_1_Tren_Prop_Vola_Mome	0	0.034502
RandomForest	B_1_Tren_Prop_Vola_Mome	1	0.035713
XGB_Aggressive	B_1_Tren_Prop_Vola_Mome	0	0.032245

Takeaway: even simple models (Ridge) can match or beat complex ones at a 1-day horizon, suggesting a very weak and unstable signal.

This motivated our pivot to direction classification.

Phase 2: Pivot to direction classification



Classification results (binary direction) vs baselines

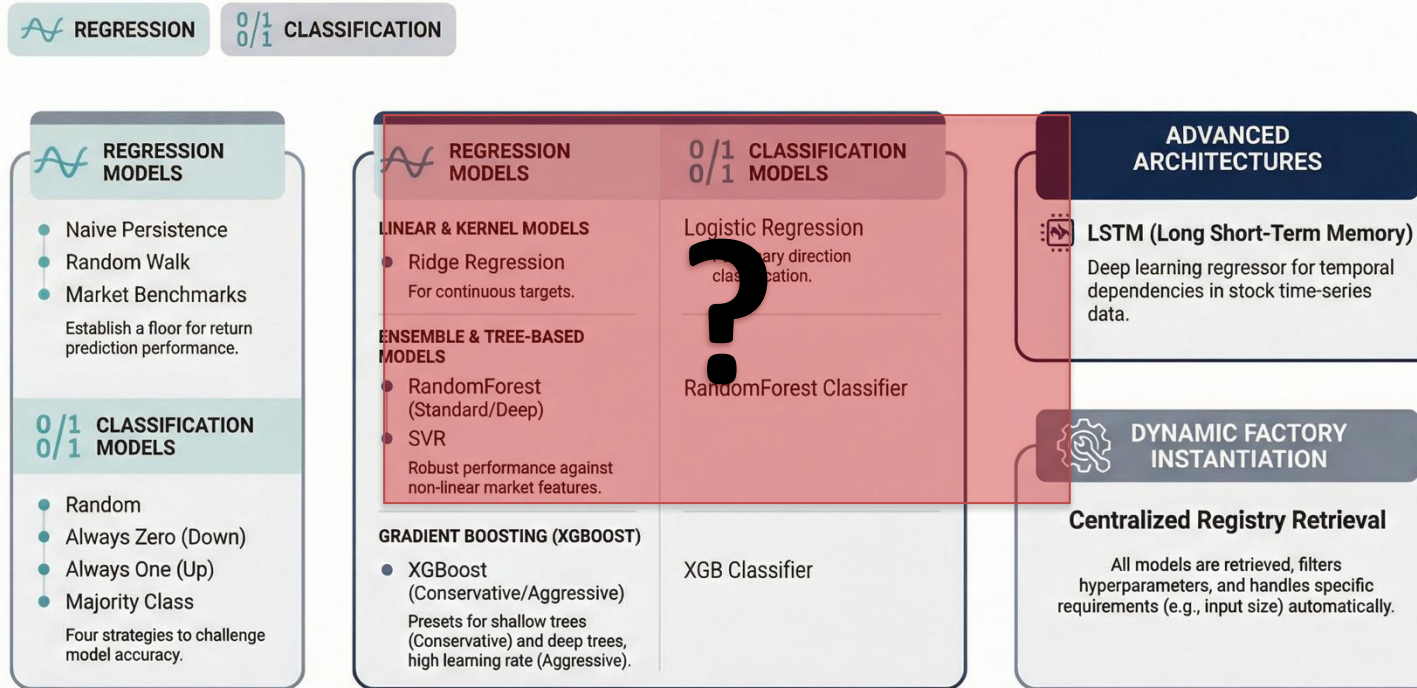
Accuracy (mean $\pm$ std) under strict temporal evaluation

Model	Best FeatureSet	Accuracy (mean $\pm$ std)
ClassificationBaselineMajor	B_11_Sent_Vola_Macr_Mome	0.528 $\pm$ 0.040
ClassificationBaselineOne	B_11_Sent_Vola_Macr_Mome	0.528 $\pm$ 0.040
ClassificationBaselineRandom	B_5_Tren_Mome	0.521 $\pm$ 0.026
LogisticRegression	B_16_Tren_Even_Vola	0.503 $\pm$ 0.045
XGBClassifier	B_13_Tren_Sent	0.501 $\pm$ 0.055
RandomForestClassifier	B_4_Tren_Even_Sent_Prop	0.492 $\pm$ 0.040
ClassificationBaselineZero	B_10_Prop_Tren_Mome	0.474 $\pm$ 0.037

Observation: strong baselines reach ~ 0.52 – 0.53 accuracy; our first-round ML models did not yet beat them consistently.

The Model Zoo: Candidate Algorithms

The Model Zoo: A Modular Factory for Predictive Architecture



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